

Persona Archetype Definitions

DA-01-T8 Feature 1 · Behavioural Segmentation · Input for synthetic dataset generation

Prepared by: DA/ML Team · T2 2026, Week 7 · For: DE team

1. Purpose

This document defines five persona archetypes to be used as target structure for a synthetic behavioural dataset. The dataset will be used to validate the DA-01 clustering pipeline end-to-end while real-user data collection continues in parallel.

Read alongside the DA-01-T7 Behavioural Data Dictionary, which specifies the behavioural features, their GA4 sources, derivations, and units.

2. Feature basis

Archetypes are defined across 12 of the 13 behavioural features specified in DA-01-T3 and DA-01-T7, spanning four dimensions: Engagement Intensity, Feature Affinity, Exploration Style, and Recency & Tenure.

Feature excluded from this synthetic run: avg_percent_scrolled. The GA4 default scroll event fires only once at 90% page depth, so the underlying signal has no variance and the feature is not usable for clustering until a custom multi-threshold scroll event is implemented on discountmate.app. Excluded from both the synthetic schema and the T2 clustering run; documented as an instrumentation gap for T3.

Persona names and thematic definitions are inherited from Adam Creek's T1 exploratory persona framework. Feature-level profiles below are defined by the DA/ML team based on plausible platform behaviour per persona.

2.1 Market alignment

Persona proportions in Section 4.1 are informed by market research on Australian grocery shoppers and users of grocery price-comparison platforms:

- Australian grocery shoppers are under sustained cost-of-living pressure: 83% cite cost of living as a major issue, 88% have noticed higher grocery prices, and 83% have taken active steps to reduce weekly expenses (ING Sense of Us 2026).
- 19.5% of Australian online shoppers have used online price comparison services, and 55% now buy groceries online at least monthly, up from 40% in 2021 (IAB Australia / Pureprofile 2026).
- Competitor positioning in the Australian grocery price-comparison space (Frugl, Bargeroo, Save On Groceries) consistently leads with three audiences: deal-seekers, families managing weekly shops, and — as a smaller named segment — health-conscious and dietary-need shoppers.

Users of price-comparison platforms self-select for price sensitivity, so proportions here are weighted toward the deal-seeking and family segments. Premium and convenience-first shoppers typically use direct retailer apps or subscription services rather than comparison tools, and are included at low proportions to reflect this.

3. Persona archetypes

Ranges are indicative bounds. DE should sample values within these ranges (uniform or Gaussian around midpoint, at DE's discretion), with appropriate noise. See Section 4 for generation constraints.

3.1 Budget-Conscious Shopper (Deal Hunter)

Price-sensitive user actively comparing products and looking for specials. Multiple return sessions with moderate depth per session.

Dimension	Feature	Expected range
Engagement	sessions_count	3 – 6
Engagement	avg_engagement_time_sec	60 – 180
Engagement	pageviews_per_session	4 – 8

Affinity	compare_ratio	0.15 – 0.35
Affinity	specials_ratio	0.15 – 0.35
Affinity	mylists_ratio	0.00 – 0.05
Affinity	category_browse_ratio	0.05 – 0.15
Affinity	product_detail_ratio	0.00 – 0.10
Exploration	unique_pages_visited	8 – 15
Exploration	browsing_entropy	0.60 – 0.85
Recency	days_since_first_visit	3 – 14
Recency	days_since_last_activity	0 – 4

3.2 Health Enthusiast

Focused reader of individual product details, typically within specific categories (health, organic). Longer engagement per session; lower interest in specials or comparison.

Dimension	Feature	Expected range
Engagement	sessions_count	2 – 4
Engagement	avg_engagement_time_sec	120 – 300
Engagement	pageviews_per_session	3 – 6
Affinity	compare_ratio	0.00 – 0.10
Affinity	specials_ratio	0.00 – 0.05
Affinity	mylists_ratio	0.00 – 0.05
Affinity	category_browse_ratio	0.10 – 0.25
Affinity	product_detail_ratio	0.15 – 0.30
Exploration	unique_pages_visited	5 – 10
Exploration	browsing_entropy	0.30 – 0.60
Recency	days_since_first_visit	3 – 14
Recency	days_since_last_activity	0 – 5

3.3 Family Planner

Broad shopper planning across many categories, with heavy use of My Lists and frequent return sessions. High page volume and wide exploration.

Dimension	Feature	Expected range
Engagement	sessions_count	4 – 8
Engagement	avg_engagement_time_sec	90 – 240
Engagement	pageviews_per_session	5 – 10
Affinity	compare_ratio	0.05 – 0.15
Affinity	specials_ratio	0.05 – 0.15
Affinity	mylists_ratio	0.15 – 0.30
Affinity	category_browse_ratio	0.10 – 0.20
Affinity	product_detail_ratio	0.05 – 0.15
Exploration	unique_pages_visited	10 – 20
Exploration	browsing_entropy	0.50 – 0.75
Recency	days_since_first_visit	5 – 14
Recency	days_since_last_activity	0 – 3

3.4 Convenience Seeker

Short, focused sessions with minimal exploration or feature use. Enters, checks a small number of pages, leaves.

Dimension	Feature	Expected range
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Engagement	sessions_count	1 – 2
Engagement	avg_engagement_time_sec	10 – 60
Engagement	pageviews_per_session	2 – 4
Affinity	compare_ratio	0.00 – 0.05
Affinity	specials_ratio	0.00 – 0.05
Affinity	mylists_ratio	0.00 – 0.05
Affinity	category_browse_ratio	0.00 – 0.10
Affinity	product_detail_ratio	0.00 – 0.05
Exploration	unique_pages_visited	2 – 5
Exploration	browsing_entropy	0.00 – 0.35
Recency	days_since_first_visit	1 – 14
Recency	days_since_last_activity	0 – 7

3.5 Premium Shopper

Reads product detail carefully but shows low sensitivity to specials or price comparison. Moderate session frequency; longer engagement.

Dimension	Feature	Expected range
Engagement	sessions_count	2 – 4
Engagement	avg_engagement_time_sec	120 – 300
Engagement	pageviews_per_session	4 – 8
Affinity	compare_ratio	0.00 – 0.10
Affinity	specials_ratio	0.00 – 0.05
Affinity	mylists_ratio	0.00 – 0.10
Affinity	category_browse_ratio	0.10 – 0.20
Affinity	product_detail_ratio	0.15 – 0.30
Exploration	unique_pages_visited	6 – 12
Exploration	browsing_entropy	0.40 – 0.70
Recency	days_since_first_visit	3 – 14
Recency	days_since_last_activity	0 – 7

3.6 Low-signal residual

Users with near-zero activity on all feature-defining behaviours. Represents drop-in and single-visit users common to any web product — included to reflect that not all eligible sessions produce a clusterable behavioural signal. Not a persona.

Dimension	Feature	Expected range
Engagement	sessions_count	1
Engagement	avg_engagement_time_sec	5 – 40
Engagement	pageviews_per_session	2 – 3
Affinity	compare_ratio	0.00
Affinity	specials_ratio	0.00
Affinity	mylists_ratio	0.00
Affinity	category_browse_ratio	0.00 – 0.05
Affinity	product_detail_ratio	0.00 – 0.05
Exploration	unique_pages_visited	2 – 3
Exploration	browsing_entropy	0.00 – 0.20
Recency	days_since_first_visit	1 – 14
Recency	days_since_last_activity	0 – 10

4. Sample data specification

4.1 Target sample size and proportions

Archetype	Proportion	Target users (of 1,500)
Budget-Conscious Shopper	35%	525
Family Planner	25%	375
Health Enthusiast	15%	225
Convenience Seeker	5%	75
Premium Shopper	5%	75
Low-signal residual	15%	225
Total	100%	1,500

Note on Premium Shopper cluster size: at 75 users, this archetype may be collapsed into Health Enthusiast during clustering, as both share long engagement time and high product_detail_ratio and differ primarily on content (not behavioural) signal. This is an expected outcome given the current feature set, not a pipeline failure — content-signal features are a documented T3 candidate.

4.2 Schema conformance

Output must match the DA-01-T7 Behavioural Data Dictionary for all included features:

- Column names, order, and data types as specified in the dictionary.
- Include the 12 behavioural features listed in Section 2 plus user_pseudo_id, reporting_window_start, reporting_window_end, is_included_in_run.
- avg_percent_scrolled is omitted from the synthetic dataset (see Section 2).
- user_pseudo_id: unique per row, string.
- is_included_in_run: 1 for all synthetic users.
- Reporting window fields: use a single 14-day window agreed with DA/ML.

4.3 Value constraints

- Ratio features (compare, specials, mylists, category_browse, product_detail): each 0-1; sum across the five per user should not exceed 0.95.
- browsing_entropy: 0-1 (Simpson Diversity Index).
- Recency fields: non-negative integers; days_since_last_activity must be less than or equal to days_since_first_visit.

4.4 Realism additions

- days_since_first_visit: introduce ~20% missing values to reflect that some eligible users had their first_visit event before the reporting window (a common GA4 pattern for returning users).
- Add mild Gaussian noise (roughly 10% of range) around sampled values within each archetype's bounds, so cluster boundaries are recoverable but not trivially perfect.

4.5 Reproducibility and deliverables

- Fixed random seed; document it in the generation script.
- Filename convention: DiscountMate_synthetic_users_YYYYMMDD.csv (must include the word "synthetic").
- Deliverables: (1) CSV file, (2) generation script or notebook, (3) one-page method note describing archetype sampling, noise model, and any deviations from this spec.

5. Contacts

Robin Le — DA/ML Cross-Team Lead (spec ownership, DE coordination)

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Noel Mathew Shillow — Data Dictionary owner (DA-01-T7)